4x4 signed analog matrix — DC to 1 MHz

LC1: 12-bit multiplying DACs; consolidated dual/quad packages.
Original Revision D is preserved in pcb/analog_matrix_4x4_rev_d.
Prototype: circuit simulation is not hardware qualification.

Inputs

File: inputs.kicad_sch

Row 0

File: row_0.kicad_sch

Row 1

File: row_1.kicad_sch

Row 2

File: row_2.kicad_sch

Row 3

File: row_3.kicad_sch

Offset Trim

File: offset_trim.kicad_sch

Control

File: control.kicad_sch

Power

File: power.kicad_sch

DAC digital and supplies

File: dac_control.kicad_sch

Sheet: /
File: analog_matrix_lc1.kicad_sch

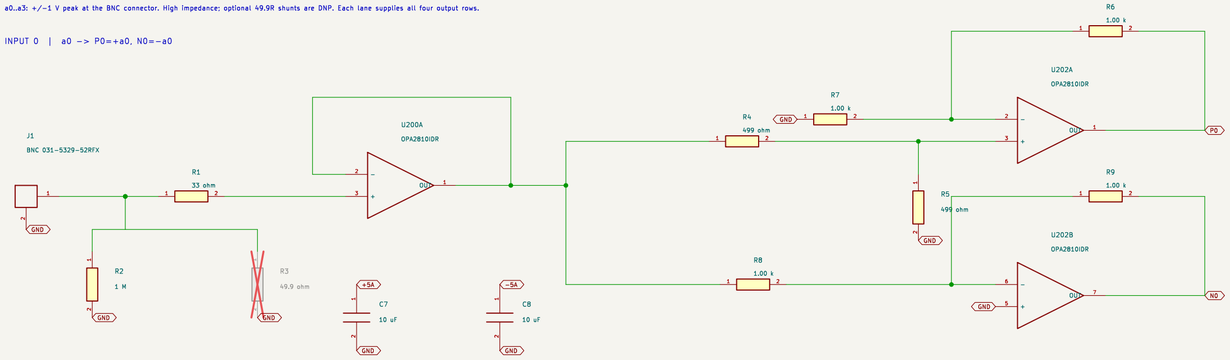
Title: Analog Programmable Matrix — LC1 lower-cost alternative

Size: A3	Date: 2026-10-02	**Rev: LC1**
KiCad E.D.A. 8.0.6		Id: 1/10

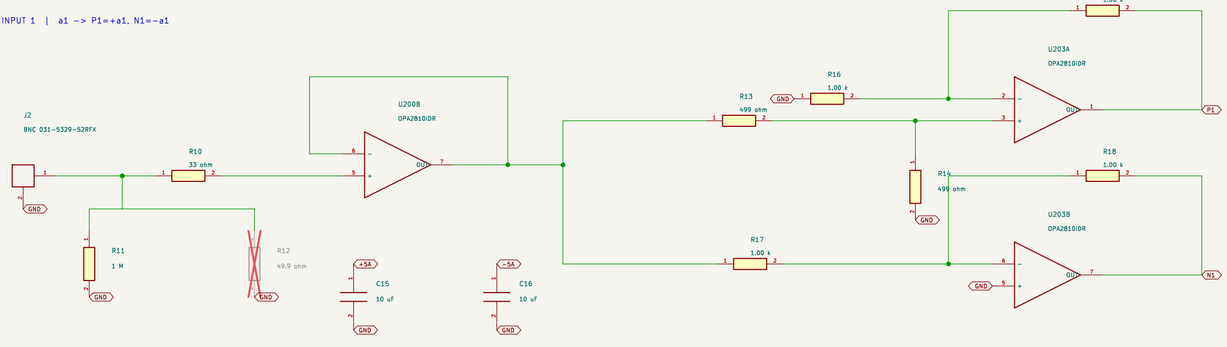
Four input buffers and matched positive / negative paths

a0..a3: +/−1 V peak at the BNC connector. High impedance; optional 49.9R shunts are DNP. Each lane supplies all four output rows.

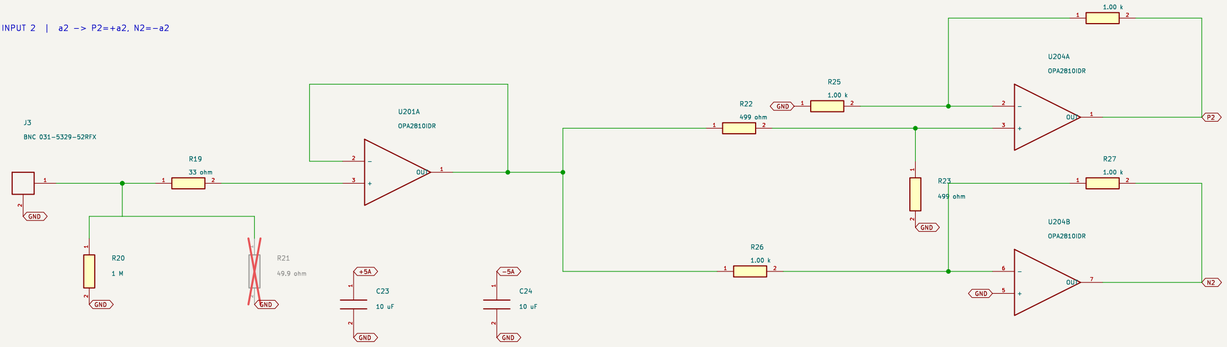
INPUT 0 | a0 -> P0=+a0, N0=−a0



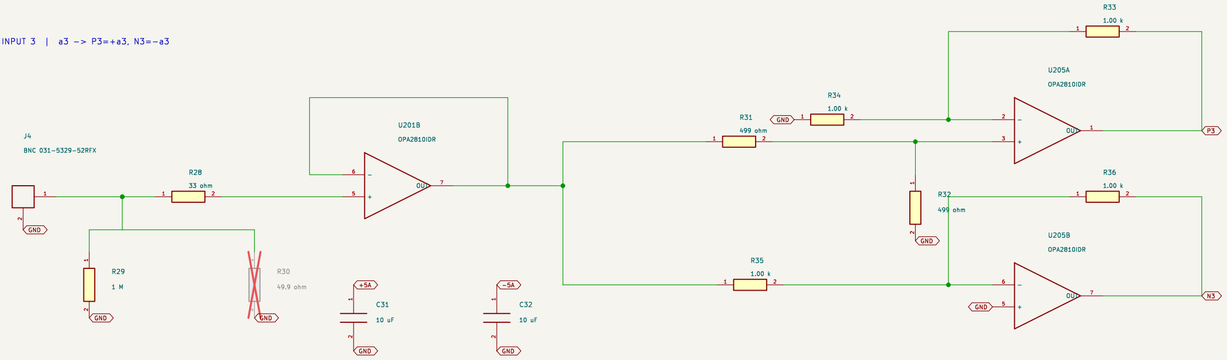
INPUT 1 | a1 -> P1=+a1, N1=−a1

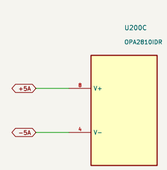


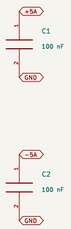
INPUT 2 | a2 -> P2=+a2, N2=−a2

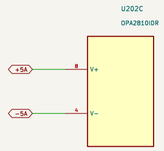


INPUT 3 | a3 -> P3=+a3, N3=−a3

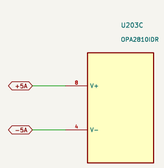




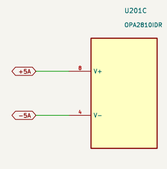




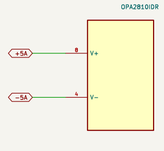




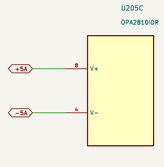














DC−1MHz; +/−1V input; +/−4V into 50 ohm. Prototype; not hardware validated.

Analog Matrix

Sheet: /Inputs/
File: inputs.kicad_sch

Title: Four input buffers and matched positive / negative paths

Size: A2	Date: 2026-10-02	**Rev: LC1**
KiCad E.D.A. 8.0.6		Id: 2/10

Output b0: signed weights and 50-ohm driver

Follow left to right: polarity -> DAC magnitude -> current-to-voltage -> sum -> line driver. Each weight includes its own local power units below.

M[0,0] | input a0

M[0,1] | input a1

M[0,2] | input a2

M[0,3] | input a3

Branch isolation: 150R for loaded-buffer stability.
TIA: 1.0pF C0G; keep IOUT and FB copper short.
DAC7811: 12-bit; digital/power units on DAC control sheet.

SUM AND OUTPUT

49.9R >=1W. External load: 50R.
+/-4V peak at the loaded connector.
No C across THS3091 feedback.

DC-1MHz; +/-1V input; +/-4V into 50 ohm. Prototype; not hardware validated.

Analog Matrix	
Sheet: /Row 0/	
File: row_0.kicad_sch	
Title: Output b0: signed weights and 50-ohm driver	
Size: A2	Date: 2026-10-02
KiCad E.D.A. 8.0.6	**Rev: LC1**
	Id: 3/10

Output b1: signed weights and 50-ohm driver

Follow left to right: polarity -> DAC magnitude -> current-to-voltage -> sum -> line driver. Each weight includes its own local power units below.

M[1,0] | input a0

M[1,1] | input a1

M[1,2] | input a2

M[1,3] | input a3

Branch isolation: 150R for loaded-buffer stability.
TIA: 1.0pF C0G; keep IOUT and FB copper short.
DAC7811: 12-bit; digital/power units on DAC control sheet.

SUM AND OUTPUT

49.9R >=1W. External load: 50R.
+/-4V peak at the loaded connector.
No C across THS3091 feedback.

DC-1MHz; +/-1V input; +/-4V into 50 ohm. Prototype; not hardware validated.

Analog Matrix	
Sheet: /Row 1/	
File: row_1.kicad_sch	
Title: Output b1: signed weights and 50-ohm driver	
Size: A2	Date: 2026-10-02
KiCad E.D.A. 8.0.6	**Rev: LC1**
	Id: 4/10

Output b2: signed weights and 50-ohm driver

Follow left to right: polarity -> DAC magnitude -> current-to-voltage -> sum -> line driver. Each weight includes its own local power units below.

M[2,0] | input a0

M[2,1] | input a1

M[2,2] | input a2

M[2,3] | input a3

Branch isolation: 150R for loaded-buffer stability.
TIA: 1.0pF C0G; keep IOUT and FB copper short.
DAC7811: 12-bit; digital/power units on DAC control sheet.

SUM AND OUTPUT

49.9R >=1W. External load: 50R.
+/-4V peak at the loaded connector.
No C across THS3091 feedback.

DC-1MHz; +/-1V input; +/-4V into 50 ohm. Prototype; not hardware validated.

Analog Matrix	
Sheet: /Row 2/	
File: row_2.kicad_sch	
Title: Output b2: signed weights and 50-ohm driver	
Size: A2	Date: 2026-10-02
KiCad E.D.A. 8.0.6	**Rev: LC1**
	Id: 5/10

Output b3: signed weights and 50-ohm driver

Follow left to right: polarity -> DAC magnitude -> current-to-voltage -> sum -> line driver. Each weight includes its own local power units below.

M[3,0] | input a0

M[3,1] | input a1

M[3,2] | input a2

M[3,3] | input a3

Branch isolation: 150R for loaded-buffer stability.
TIA: 1.0pF C0G; keep IOUT and FB copper short.
DAC7811: 12-bit; digital/power units on DAC control sheet.

SUM AND OUTPUT

49.9R >=1W. External load: 50R.
+/-4V peak at the loaded connector.
No C across THS3091 feedback.

DC-1MHz; +/-1V input; +/-4V into 50 ohm. Prototype; not hardware validated.

Analog Matrix	
Sheet: /Row_3/	
File: row_3.kicad_sch	
Title: Output b3: signed weights and 50-ohm driver	
Size: A2	Date: 2026-10-02
KiCad E.D.A. 8.0.6	**Rev: LC1**
	Id: 6/10

DC OFFSET TRIM — I2C, 12-bit, filtered

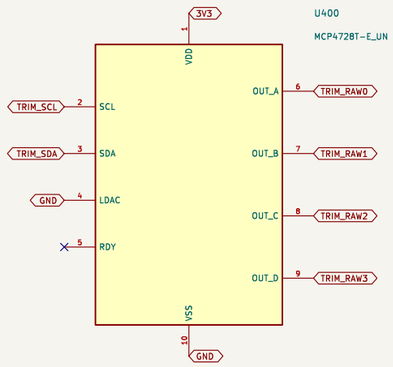
Internal reference 2.048 V; gain 1. Code 2500 gives nominal −1.250 V trim.
10k/10n feedback filters DAC updates. Keep outputs muted for 2 ms after changes.

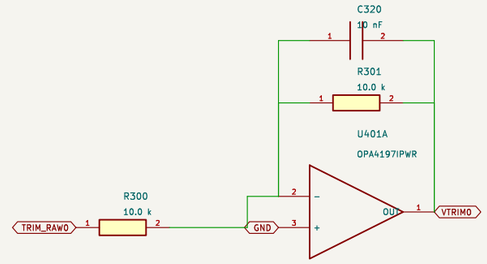
Row 0

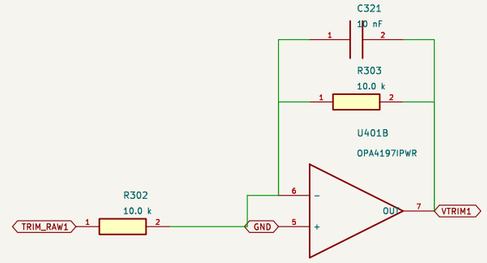
Row 1

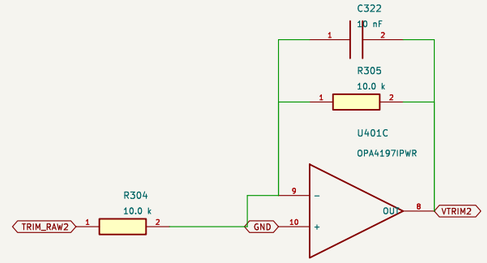
Row 2

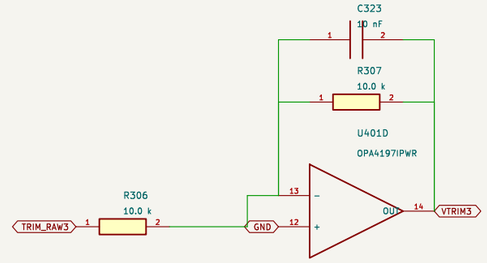
Row 3

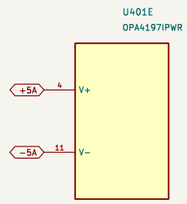
















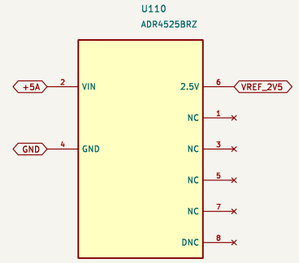








2.5 V SUMMING BIAS











Separate 12-bit prototype; see simulation model scope and bring-up requirements.

Analog Programmable Matrix

Sheet: /Offset Trim/
File: offset_trim.kicad_sch

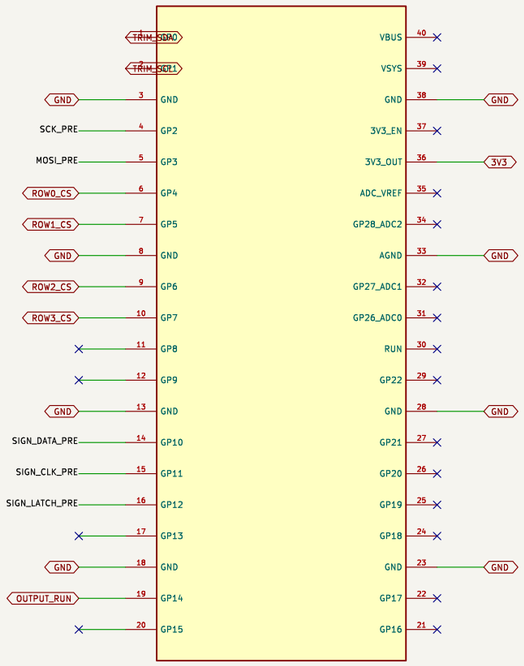
Title: LC1: DC offset correction and precision bias

Size: A3	Date: 2026-10-02	**Rev: LC1**
KiCad E.D.A. 8.0.6		Id: 7/10

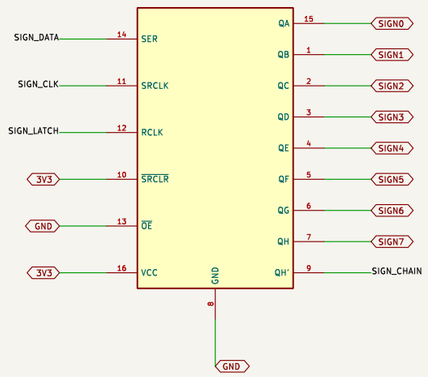
LC1 USB CONTROLLER — 12-bit coefficients and I2C offset trim

Pico is USB-powered. Do not connect VBUS/VSYS to +5A. Apply analog power first. GP14=OUTPUT_RUN defaults low.

U120
Raspberry Pi Pico

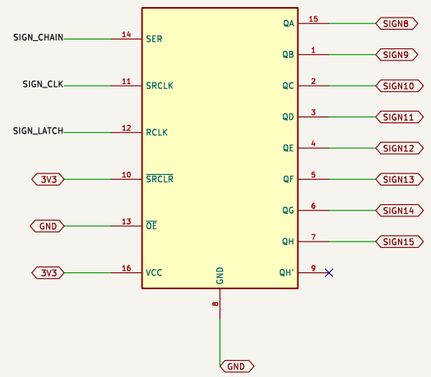


U121
SN74HC595D



First register QA..QH = SIGN0..7.
Second register QA..QH = SIGN8..15.
Clock bit 15 first, latch after 16 clocks.

U122
SN74HC595D

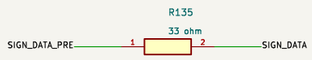


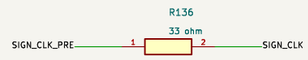


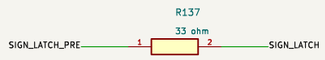








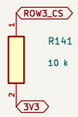














SPI mode 1; 1 MHz. Column 3 first, column 0 last.
GP0/GP1: I2C trim. GP14: output run; outputs default muted.
Allow 2 ms after trim changes before unmuting.

DC–1MHz; +/–1V input; +/–4V into 50 ohm. Prototype; not hardware validated.

Analog Matrix

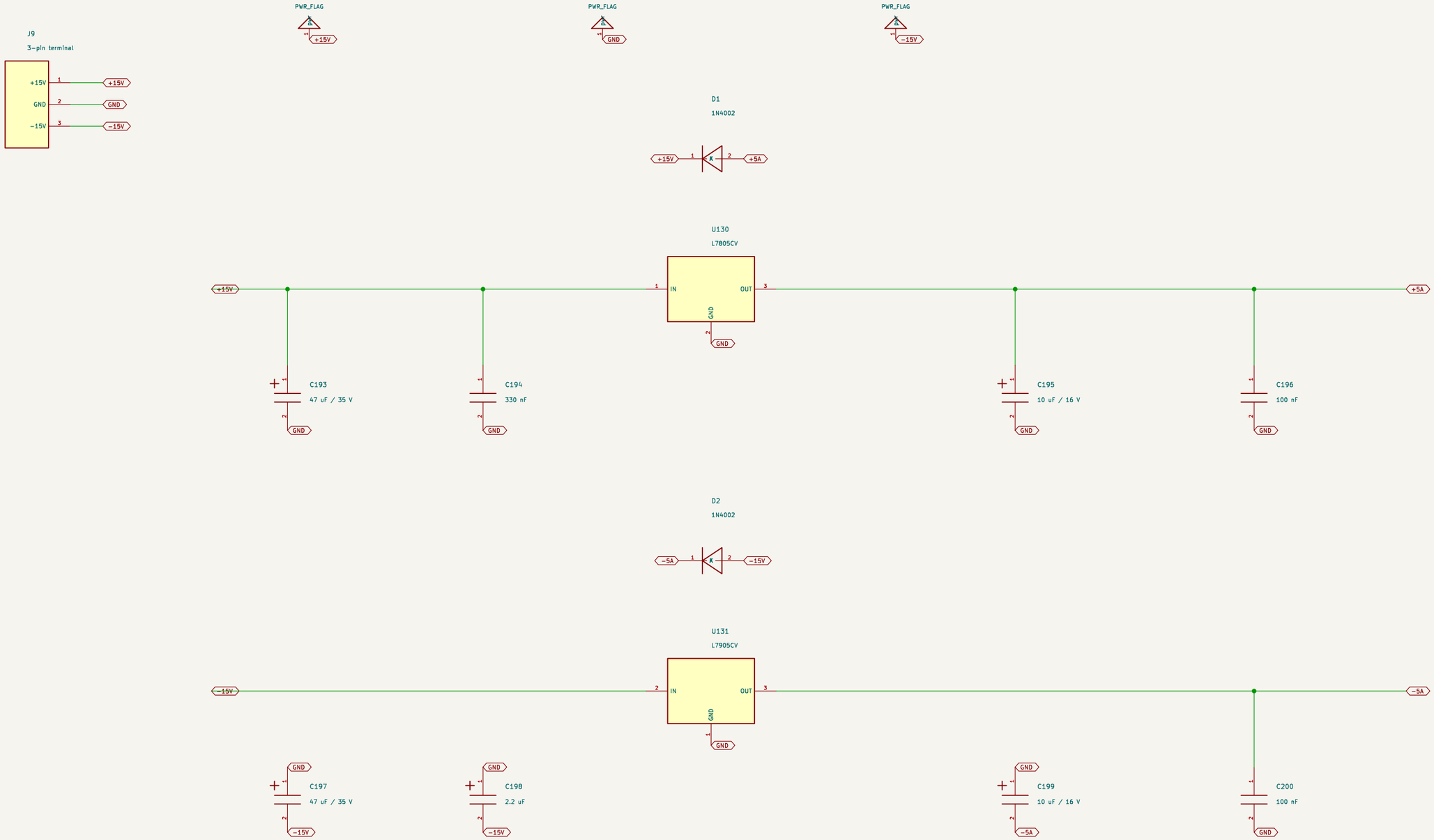
Sheet: /Control/
File: control.kicad_sch

Title: RP2040 USB / SPI controller and sign-bit latches

Size: A3	Date: 2026-10-02	**Rev: LC1**
KiCad E.D.A. 8.0.6		Id: 8/10

External +/−15 V input and +/−5 V analog regulators

External regulated +/−15 V, at least 1 A per rail. Regulator heatsinks <=15 C/W. Negative regulator tab is −15 V; positive regulator tab is GND.



Observe all polarized capacitor signs. Use the shown electrolytic/tantalum regulator capacitors; do not substitute ceramic-only outputs. Regulator dissipation is roughly 3 W per device; verify temperature under sustained DC output loads. +/−5A supplies the analog core.

DC–1MHz; +/−1V input; +/−4V into 50 ohm. Prototype; not hardware validated.

Analog Matrix

Sheet: /Power/
File: power.kicad_sch

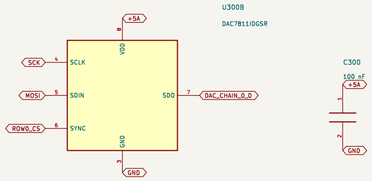
Title: External +/−15 V input and +/−5 V analog regulators

Size: A3	Date: 2026-10-02	**Rev: LC1**
KiCad E.D.A. 8.0.6		Id: 9/10

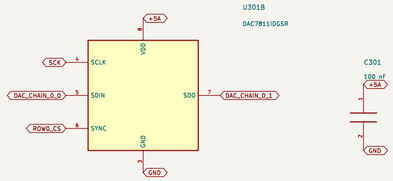
DAC7811 — FOUR INDEPENDENT ROW CHAINS

5 V DAC supplies; 3.3 V SPI inputs meet VIH. SDO is 5 V: connect only to the next DAC.
SPI mode 1, 1 MHz. Send column 3 first, column 0 last; each SYNC rising edge updates its row.
Each 100 nF bypass must sit at its DAC supply pins. Analog units are on their output—row sheets.

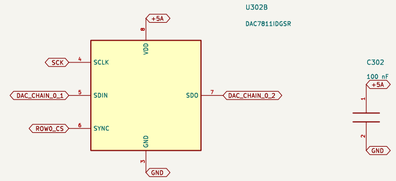
ROW 0 / COLUMN 0



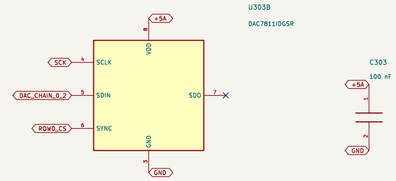
ROW 0 / COLUMN 1



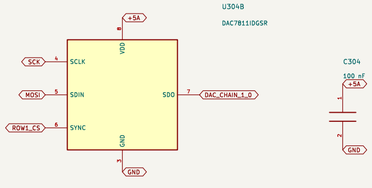
ROW 0 / COLUMN 2



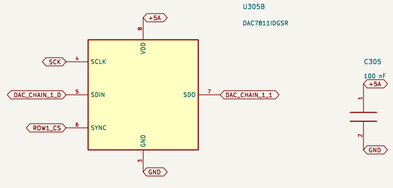
ROW 0 / COLUMN 3



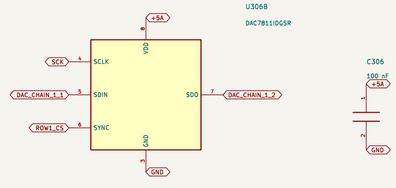
ROW 1 / COLUMN 0



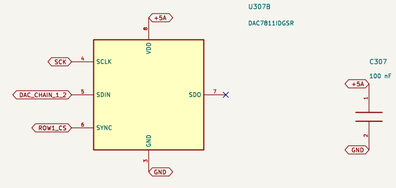
ROW 1 / COLUMN 1



ROW 1 / COLUMN 2



ROW 1 / COLUMN 3



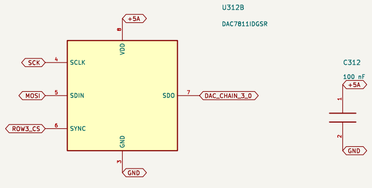
ROW 2 / COLUMN 0

ROW 2 / COLUMN 1

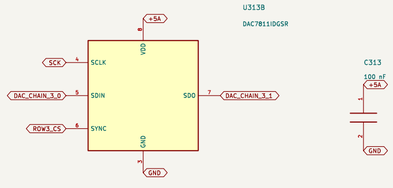
ROW 2 / COLUMN 2

ROW 2 / COLUMN 3

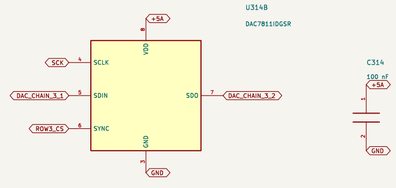
ROW 3 / COLUMN 0



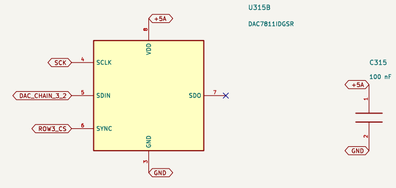
ROW 3 / COLUMN 1



ROW 3 / COLUMN 2



ROW 3 / COLUMN 3



Separate 12-bit prototype; see simulation model scope and bring-up requirements.

Analog Programmable Matrix

Sheet: /DAC digital and supplies/
File: dac_control.kicad_sch

Title: LC1: DAC row chains, power and local bypass

Size: A2	Date: 2026-10-02	**Rev: LC1**
KiCad E.D.A. 8.0.6		Id: 9/10